

OVERVIEW

Senior software engineer and acting tech lead with experience leading cross-functional infrastructure, cloud, and backend initiatives across Atlassian, Starburst, and high-growth startups. Delivered \$1M+ in annualized support-cost savings, led small engineering teams, and owned migrations, reliability improvements, and LLM-enabled internal tools. Strong in Java, TypeScript, Haskell, PostgreSQL, Kubernetes, cloud infrastructure, and complex legacy systems.

TECHNICAL SKILLS

Java, Haskell, TypeScript, PostgreSQL, Agile, Python, Perl, R, Linux, Bash, STAN, Kubernetes, Helm, Pulumi, TLS, TLA+, Cloud Networking, GitOps for CI/CD, DataDog, Svelte, React, AWS, GCP, Azure.

ADVANCED ACADEMIC SKILLS

Type Theory, Compilers, Formal Verification, Distributed Systems Analysis, Machine Learning, Bayesian Analysis, Linear Algebra, Differential Equations, Statistics

EXPERIENCE

Atlassian, Boston, MA • Senior Software Engineer & Acting Tech Lead	03/2024 - current
- Acting tech lead for error-code standardization initiative projected to deliver ~\$2M in annual support-cost savings. - Led development of support-engineering features that made error codes more discoverable, actionable, and auditable. - Led a 4-person engineering team building automated code-migration tooling across service repositories. - Owned product, architecture, implementation planning, and stakeholder alignment across engineering, design, and platform teams. - Served as team SME for LLM-assisted engineering workflows, evaluation, and internal support tooling.	
Starburst Data, Boston, MA • Senior Software Engineer & Tech Lead	10/2022 - 08/2023
- Tech Lead on a cloud infrastructure team, managing vendor relationships (AWS/GCP/Azure), cost reporting (Vantage), security (SOC2) compliance, and Cloudflare integration. - Software lead for zero downtime migration for a new domain using Cloudflare Tunnels, increasing scalability by avoiding costs associated with NAT Gateways, and taking ownership of Cloudflare management of DNS and TLS features for deployment. - TLS Certificate Re-use Project: designed and led a step by step migration to reduce deployment time by re-using TLS certificates between blue-green deployments, making deployments ~20% faster with 50% reduction in time to detection of failures. - Oversaw team of 3 software engineers making changes to Java codebase and GitOps scripts.	
SimSpace, Boston, MA • Senior Software Engineer & Tech Lead	12/2019 - 09/2022

- Tech Lead on security observability tools for cyber ranges using TypeScript, NodeJS, Kubernetes and systems design for data pipeline.
- Built backend services for learning management systems using Haskell and PostgreSQL. Extended a >500k LOC Haskell project, added capabilities for end users, correctness tests, and load tests.
- Designed and implemented a type safe continuation token paging feature within our internal Haskell pagination library for better user experience and performance..

College Vine, Cambridge, MA • Software Engineer & Data Scientist

08/2017 - 09/2019

- Led development of machine learning and predictive algorithms used in production by thousands of end users to estimate a college applicant's chances of admission.
- Several Ad Hoc Funnel metrics: creation of a bespoke lead score model for HubSpot using Bayesian feature selection and linear regression.
- Built a "decision support" app using Python and React so students could make optimal choices on which classes prepare them best for their college aspirations.

Elsen, Boston, MA • Junior Software Engineer & Data Engineer

10/2015 - 08/2017

- Created a data pipeline responsible for implementing data ingress, responsible for ingress of up to data financial data from several different sources.
- Enhanced in house programming language used for exploratory analysis of financial data, backtesting of trading strategies, and improving performance for to end users.

Umass Medical School, Worcester, MA • Bioinformatician

01/2011 - 12/2014

- Conducted genomic research on the human genome, protein structure, and human computer interaction as a member of several labs, whose work directly led to five publications and contributed to several others. [Link](#).
- Created web based interfaces used by scientists for the visualization of research results, leading to figures in prominent publications (Nature).

EDUCATION AND AWARDS

Georgia Tech, Atlanta, GA • Master's in Computer Science

Graduation Year (2021)

GPA: 3.9

University of Vermont, Burlington, VT • Bachelor's of Biological Science

Graduation Year (2010)

Knuth Check #1660

2019

Eagle Scout Award

2005

OPEN SOURCE

Maintainer of two open source projects in the Haskell ecosystem, [Repline](#) and [Protolude](#).

Writer of Write Yourself a Scheme in 48 Hours, Version 2.0 — [Github](#)

Founder of Red Book Reading Group - [Discord Group](#) for weekly reading of [redbook.io](#)